Thyroid *and* antithyroid drugs

Pharmacology • Therapeutics • Monitoring • Safety

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LEARNING OBJECTIVES

Objectives

นักศึกษาสามารถ...

01

DRUG CLASSES

บอกชื่อ thyroid drugs และ anti-thyroid drugs ได้

02

MECHANISM

อธิบายกลไกการออกฤทธิ์และฤทธิ์ทางเภสัชวิทยา

03

CLINICAL USE

อธิบายประโยชน์ทางคลินิก

04

SAFETY

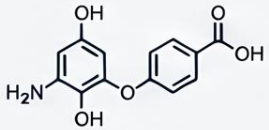
อธิบายอาการไม่พึงประสงค์ ข้อควรระวัง และข้อห้ามใช้

Comparative Pharmacology of Thyroid Drugs

A high-level comparative overview of thyroid medications for clinical decision-making, focusing on pharmacokinetics, indications, and safety monitoring.

Thyroid Replacement Therapy (T4 vs. T3)

Levothyroxine (T4) Standard of Care



Long Half-Life
(7 days)

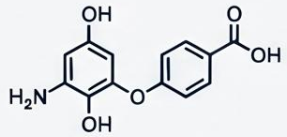


Stable Protein Binding
Consistent physiological levels.



Critical Administration Timing
Must be taken on an **empty stomach** to avoid absorption interactions with food or minerals.

Liothyronine (T3) Rapid Onset



Short Half-Life
(frequent dosing)

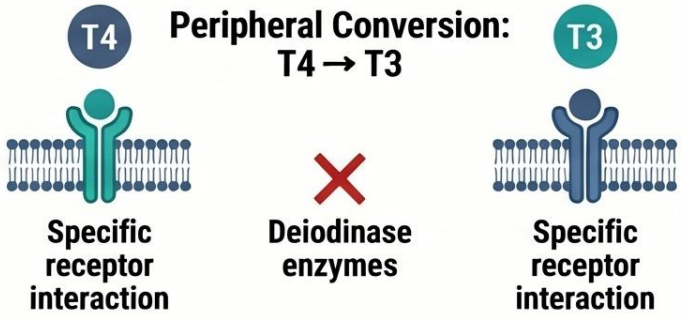


Rapid Effect
Used for rapid effect or in combination.

Thyroid Hormone Synthesis & Peripheral Conversion Pathway

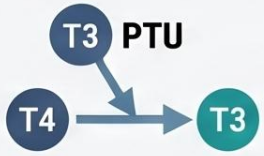


Peripheral Conversion: T4 → T3



Antithyroid Pharmacotherapy (Thionamides)

PTU's Peripheral T4 to T3 Inhibition



Unlike others, **PTU inhibits peripheral conversion,** making it vital for thyroid storm management.

Carbimazole as a Prodrug

Carbimazole

Enzymatic process

Methimazole
(active metabolite)

Rapidly converted to **Methimazole** after administration.

Trimester-Specific Pregnancy Guidance



First Trimester
PTU preferred.



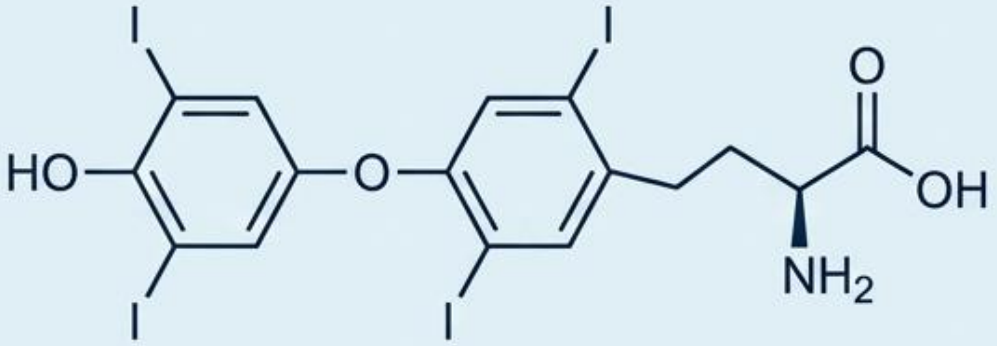
Second & Third Trimesters
Methimazole recommended.

Primary Monitoring & Adverse Effects

Drug Name	Mechanism of Action	Half-Life	Primary Monitoring & Safety Issues
Levothyroxine	Synthetic T4 (Prohormone)	~7 Days	**TSH, Free T4. Warning:** Boxed issues not common at therapeutic doses.
Methimazole	Inhibits Thyroid Peroxidase	5–6 Hours	**TSH, LFTs, CBC. Warning:** Agranulocytosis risk.
PTU	Inhibits TPO + Peripheral T4/T3 conversion	1–2 Hours	**Liver Function** (Hepatotoxicity risk). **Warning:** Severe liver injury risk.

Evidence-based references: Endocrine Society Clinical Guidelines / Pharmacology Testbooks.

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T4 (Levothyroxine)

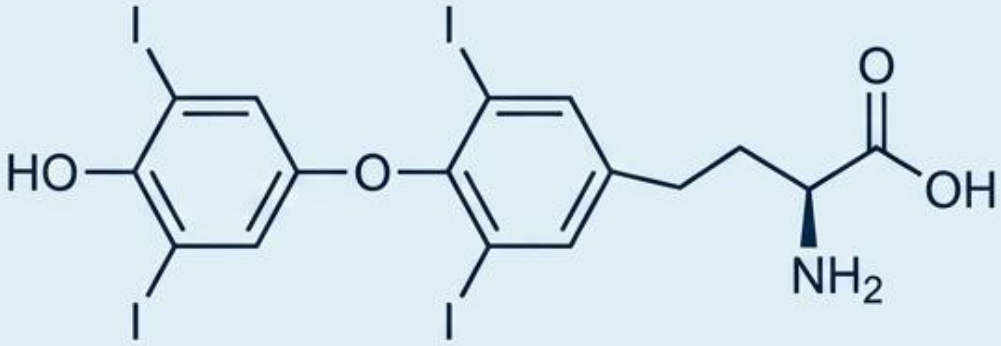
Biologic Potency: 1x (Prohormone)

Half-life: 7 days

Volume of Distribution: 10 L

Daily Production: 75 mcg

Fractional Turnover: 10% / day



T3 (Liothyronine)

Biologic Potency: 4x (Biologically active form)

Half-life: 1 day

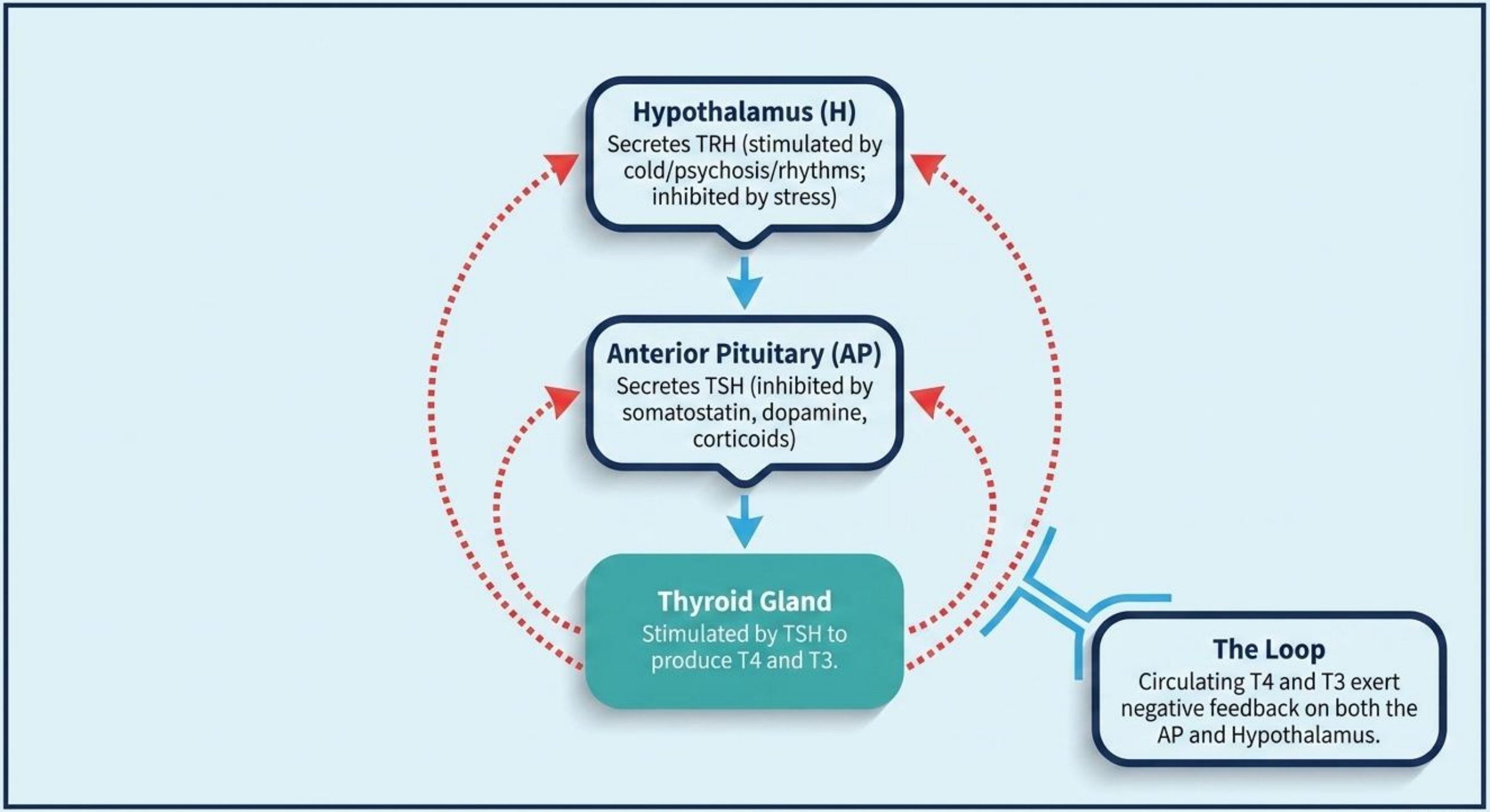
Volume of Distribution: 40 L

Daily Production: 25 mcg

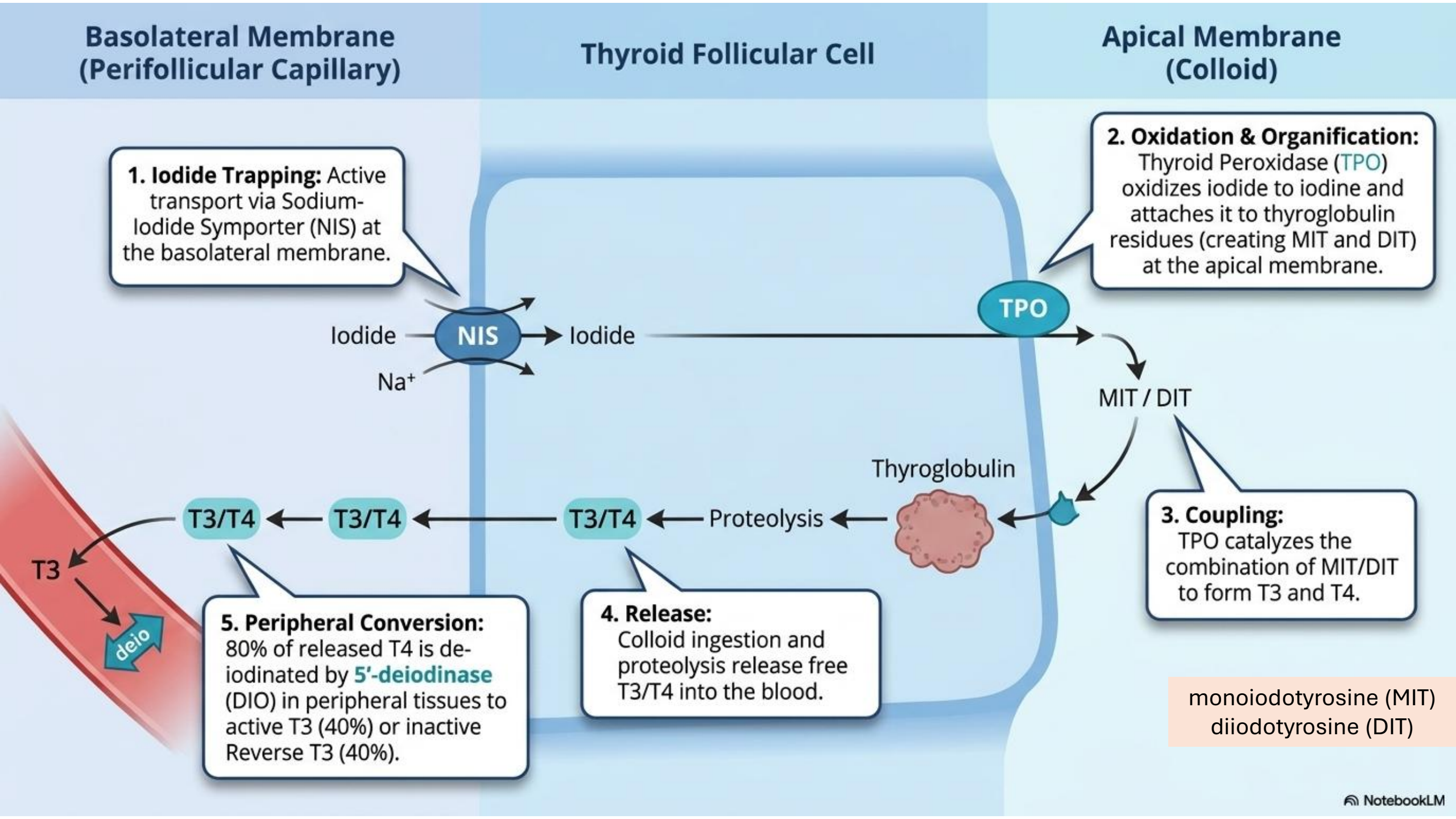
Fractional Turnover: 60% / day

Protein Binding Note: Both are highly bound (>99%), primarily to Thyroxine-Binding Globulin (TBG), Transthyretin, and Albumin.

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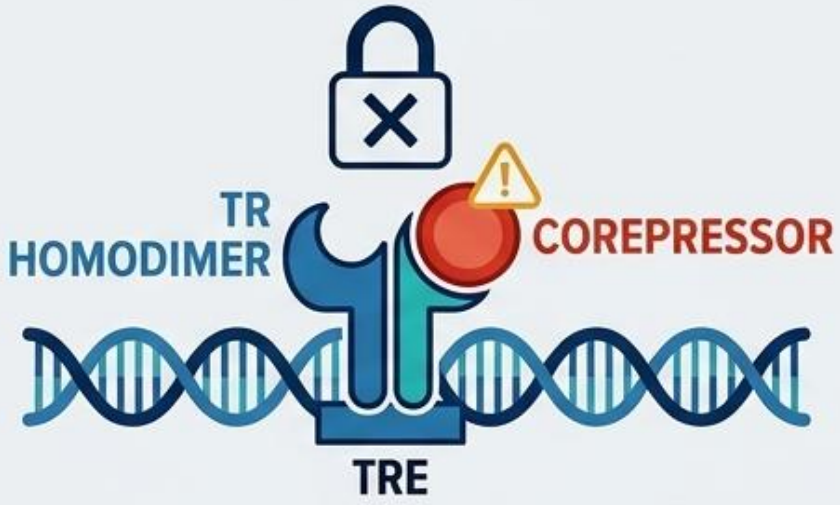

Hypothalamus (H)
Secretes TRH (stimulated by cold/psychosis/rhythms; inhibited by stress)
Anterior Pituitary (AP)
Secretes TSH (inhibited by somatostatin, dopamine, corticoids)
Thyroid Gland
Stimulated by TSH to produce T4 and T3.
The Loop
Circulating T4 and T3 exert negative feedback on both the AP and Hypothalamus.

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Basolateral Membrane (Perifollicular Capillary)
Thyroid Follicular Cell
Apical Membrane (Colloid)
1. Iodide Trapping: Active transport via Sodium-Iodide Symporter (NIS) at the basolateral membrane.
2. Oxidation & Organification: Thyroid Peroxidase (TPO) oxidizes iodide to iodine and attaches it to thyroglobulin residues (creating MIT and DIT) at the apical membrane.
Iodide
NIS
Iodide
Na+
TPO
MIT / DIT
Thyroglobulin
T3/T4
T3/T4
T3/T4
Proteolysis
T3
deio
3. Coupling: TPO catalyzes the combination of MIT/DIT to form T3 and T4.
4. Release: Colloid ingestion and proteolysis release free T3/T4 into the blood.
5. Peripheral Conversion: 80% of released T4 is de-iodinated by 5'-deiodinase (DIO) in peripheral tissues to active T3 (40%) or inactive Reverse T3 (40%).
monoiodotyrosine (MIT)
diiodotyrosine (DIT)
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T3 ACTS AS A NUCLEAR KEY TO ALTER GENE TRANSCRIPTION

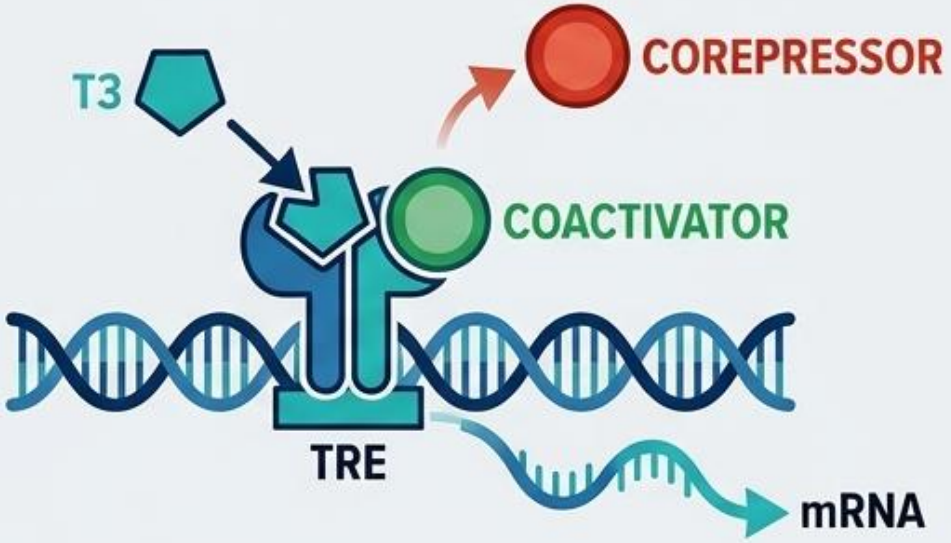
INACTIVE STATE (TRANSCRIPTION REPRESSED)



Unliganded T3 receptor (TR) binds to the Thyroid Response Element (TRE) with a corepressor. Transcription is blocked.

T3 ENTERS NUCLEUS

ACTIVE STATE (TRANSCRIPTION PROMOTED)



T3 binds TR → corepressor is displaced → coactivator attaches → gene transcription alters cellular phenotype.

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HYPOTHYROIDISM

From cause to treatment: a single clinical pathway

Reduced thyroid hormone slows metabolism across multiple organ systems.

1 CAUSE

- Autoimmune thyroid failure (Hashimoto disease)
- Thyroiditis or congenital disease
- Thyroidectomy / radioiodine
- Iodine imbalance or medicines

Central cause: pituitary / hypothalamic failure

2 CONFIRM

Thyroid hormone deficiency

PRIMARY PATTERN
↑ TSH + ↓ free T4

CENTRAL PATTERN
Low or inappropriately normal TSH + ↓ free T4

Symptoms alone are not diagnostic

3 CLINICAL EFFECT

- Fatigue / cold intolerance
- Weight gain / constipation
- Dry skin / bradycardia
- Slowed cognition / depression

Severe: hypothermia, respiratory depression, reduced consciousness

Myxedema is a medical emergency

4 TREAT

LEVOTHYROXINE (T4)
First-line replacement

MONITOR
TSH in primary disease
Free T4 in central disease

GOAL
Normalize labs + relieve symptoms

Liothyronine (T3) is not routine

CLINICAL LOGIC

Loss of thyroid hormone → multisystem slowing → replace T4 and monitor response

Myxedema is a term used synonymously with severe hypothyroidism, but also to describe a dermatological change that can occur in hypothyroidism and (rare) paradoxical cases of hyperthyroidism.

LEVOTHYROXINE REPLENISHES THE SYSTEMIC RESERVOIR

T4

MECHANISM OF ACTION

- **Synthetic T4 prohormone.**
- Relies entirely on intracellular conversion to T3.
- Renders exogenous T3 administration unnecessary.

PHARMACOKINETICS

- Prolonged **7-day half-life.**
- **Requires 4-5 weeks** to reach steady-state plasma concentrations.
- Absorption is incomplete (70-80%).

CLINICAL INDICATIONS

- **Gold standard** for hypothyroidism replacement.
- Average adult replacement dose: **1.7 µg/kg**.
- Used for **TSH suppression** in thyroid cancer.

INTERACTIONS & ADMINISTRATION

- Must be taken on an **strictly empty stomach**.
- Absorption impaired by **Iron, Calcium,** and **antacids**.
- **Metabolism accelerated** by **CYP3A4 inducers** (rifampicin, phenytoin).

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Levothyroxine: dose, monitoring and safety

Initial dose

Healthy younger adults: full replacement is about 1.6 micrograms/kg/day. Individualize the dose.

Older adults / cardiac disease

Start lower, often 12.5–25 micrograms/day, and titrate cautiously to avoid angina or arrhythmia.

Follow-up

Primary disease: check TSH about 6–8 weeks after a dose change. Central disease: follow free T4.

Over-replacement

Can cause palpitations, atrial fibrillation and bone loss. Treat adrenal insufficiency before thyroid hormone.

Pregnancy often increases the T4 requirement.

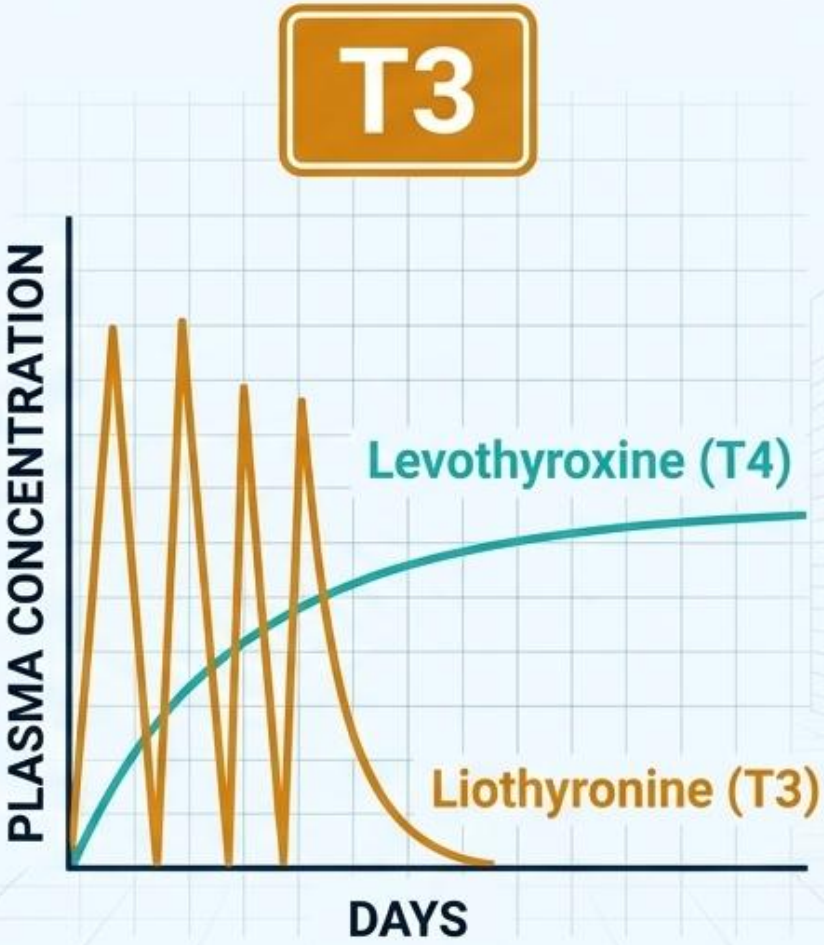
LIOTHYRONINE PROVIDES RAPID REVERSAL IN ACUTE CRISES

MECHANISM OF ACTION

- Synthetic active T3.
- 3-4x more potent than Levothyroxine.

CLINICAL INDICATIONS

- Rarely used for routine maintenance.
- Reserved strictly for acute emergencies (e.g., Myxedema Coma) requiring immediate physiological reversal.



PHARMACOKINETICS

- Short 1-day half-life.
- Extremely rapid onset of action.
- Volatile plasma levels require multiple daily doses.

CAUTION & LIMITATIONS

- High risk of iatrogenic cardiotoxicity.
- Avoid entirely in patients with underlying cardiac disease.
- Difficult to monitor adequacy via standard laboratory tests.

Iatrogenic cardiotoxicity refers to heart damage or dysfunction caused directly by medical treatments, drugs, or interventions prescribed for other conditions

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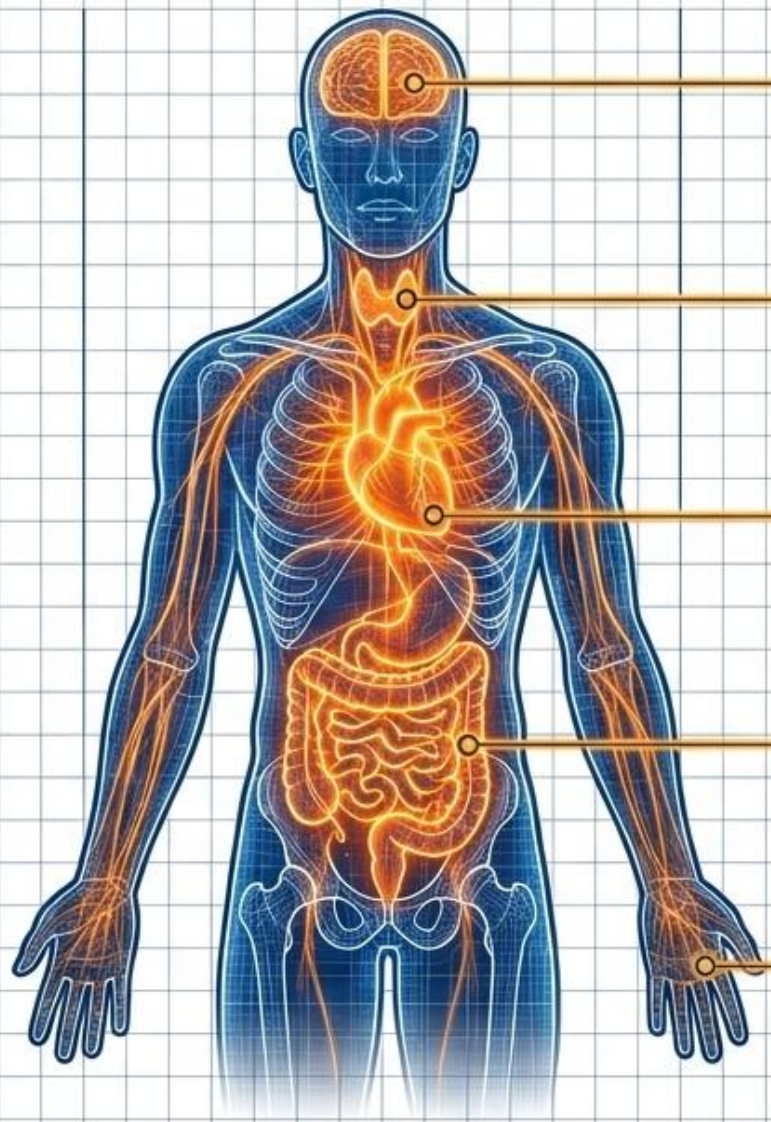
HYPERTHYROIDISM ACCELERATES SYSTEMIC ACTIVITY

PATHOLOGY

- Elevated free thyroid hormones (Thyrotoxicosis).
- Often driven by TSH receptor autoantibodies (Graves' disease).

THERAPEUTIC GOALS

- 1. Eliminate excessive hormone production.
- 2. Control systemic hypermetabolic symptoms.



- Nervousness, irritability, insomnia
- Enlarged thyroid gland (Goiter)
- Tachycardia, palpitations, arrhythmias
- Weight loss (despite hunger), diarrhea
- Tremor, heat sensitivity, sweating

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HYPERTHYROIDISM
Treatment Options

Goal: Restore euthyroid state and relieve symptoms

1 ANTITHYROID DRUGS



- Inhibit synthesis of thyroid hormones
- First-line therapy for Graves' disease
- May induce remission

Examples

- Methimazole
- Propylthiouracil (PTU)
- Carbimazole

Adjunctive therapy

- Beta-blockers
- Symptom relief (e.g. propranolol)

2 RADIOACTIVE IODINE (RAI)

- I-131 destroys overactive thyroid tissue
- Non-surgical, effective definitive therapy
- May lead to hypothyroidism (monitor and treat)

Safe and well-tolerated

Outpatient procedure

Avoid pregnancy for 6–12 months

3 SURGERY



- Removal of part or all of the thyroid gland
- Rapid control of hyperthyroidism
- Preferred in specific situations

Consider when:

- Large goiter causing compression
- Suspicious or coexisting thyroid cancer
- Contraindication or intolerance to drugs / RAI
- Patient preference

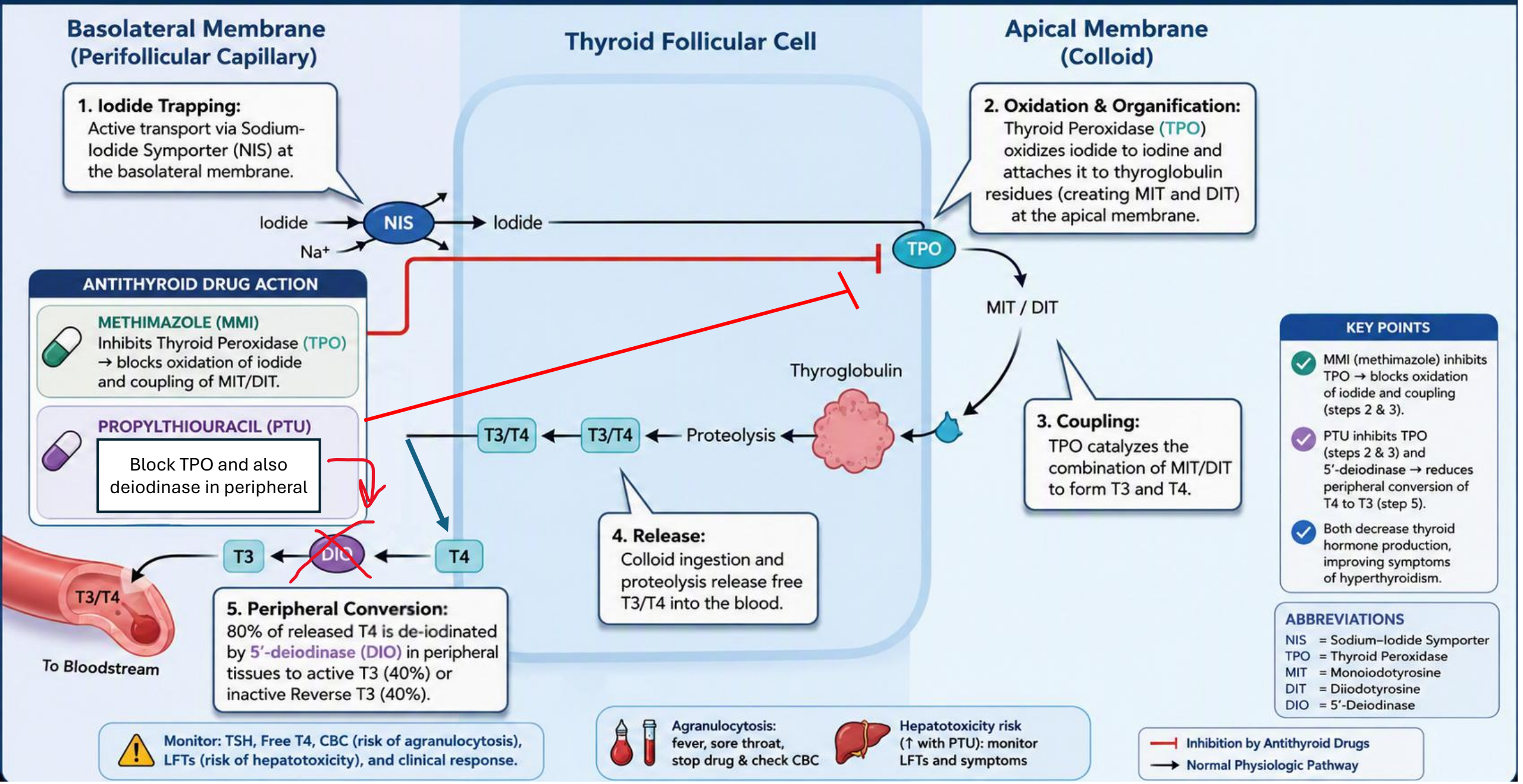
GOALS OF THERAPY

- Eliminate excessive thyroid hormone production
- Control the symptoms of hyperthyroidism

Treatment choice depends on patient factors, severity, and preference!

Monitor regularly: TSH, Free T4/T3, clinical response, and adverse effects

THYROID HORMONE SYNTHESIS AND MECHANISM OF ANTITHYROID DRUGS



Thioamides: MMI and PTU

Property	Methimazole (MMI)	Propylthiouracil (PTU)
Thyroid peroxidase	**Inhibits organification and coupling**	**Inhibits organification and coupling**
Peripheral T4-to-T3 conversion	**No clinically important inhibition**	**Inhibits type 1 deiodinase (D1)**
Typical dosing frequency	Usually once daily	Usually divided doses
Usual place in therapy	Preferred in most nonpregnant adults	Selected indications
Important risks	Agranulocytosis, liver injury, embryopathy	Agranulocytosis, severe liver injury, vasculitis

Carbimazole converts to MMI. Both drugs reduce new hormone synthesis.

Thioamides: adverse effects and patient advice

Problem	Key point	Patient action
Agranulocytosis	Rare, potentially life-threatening. Often early, but may occur later	Fever OR sore throat: stop drug and obtain urgent CBC with differential
Liver injury	PTU: severe hepatocellular injury. MMI: often cholestatic injury	Jaundice, dark urine or marked abdominal symptoms: stop and seek urgent assessment
ANCA-associated vasculitis	More strongly associated with PTU	New purpura, hematuria or respiratory symptoms need assessment
Rash, arthralgia, GI upset	More common, often less severe	Contact the treating clinician for advice

ANCA = antineutrophil cytoplasmic antibodies. ANCA-associated vasculitis (AAV) is a group of rare, systemic autoimmune diseases characterized by inflammation and destruction of small blood vessels (capillaries, venules, and arterioles).

After agranulocytosis, avoid switching to the other thioamide because cross-reactions can occur.

Thioamides: Risk Profile

Agranulocytosis (The Most Serious)



0.1-0.5% incidence. Usually occurs within the first 90 days.
PTU = idiosyncratic;
MMI = dose-related.
Patient **must stop drug immediately** if they develop sore throat + fever.

Hepatotoxicity



PTU = Severe hepatocellular injury/hepatitis.
MMI = Cholestatic jaundice.

Vasculitis



Drug-induced ANCA-associated vasculitis (PTU > MMI).



Common/Minor ADRs: Maculopapular pruritic rash (most common, 4-6%), GI upset, arthralgia. (Note: 50% cross-sensitivity between PTU and MMI).

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Monitoring antithyroid treatment

Before treatment

Check free T4/T3 and TSH, CBC with differential, and a liver profile. Discuss pregnancy and warning symptoms.

Early follow-up

Recheck free T4 and T3 in about 2–6 weeks. Adjust dose to hormone levels and clinical response.

TSH may lag

TSH can remain suppressed for months. Early dose decisions rely mainly on free T4 and T3.

Safety during treatment

Fever or sore throat needs urgent CBC. Symptoms of liver injury need prompt liver tests and assessment.

A normal previous CBC does not remove the need to act on new warning symptoms.

Thioamides: why the effect takes time

New synthesis falls

MMI and PTU inhibit TPO in the thyroid. Existing colloid stores remain available for release.

Circulating hormone persists

T4 has a long half-life. Clinical and biochemical improvement therefore develops over weeks.

PTU adds peripheral action

D1 inhibition reduces some T4-to-T3 conversion. It does not remove stored thyroid hormone.

Symptoms need a bridge

A beta-blocker can relieve palpitations and tremor while antithyroid treatment takes effect.

Thioamides are ineffective when excess hormone comes from destructive thyroiditis.

Thioamides: choosing MMI vs PTU

01

Graves’ disease / increased synthesis

MMI is preferred for most nonpregnant adults. It usually allows once-daily dosing. Hormone levels fall over several weeks.

02

Thyroid storm (thyrotoxic crisis)

PTU also inhibits T4-to-T3 conversion. MMI is an effective alternative. The emergency protocol and liver status guide selection.

03

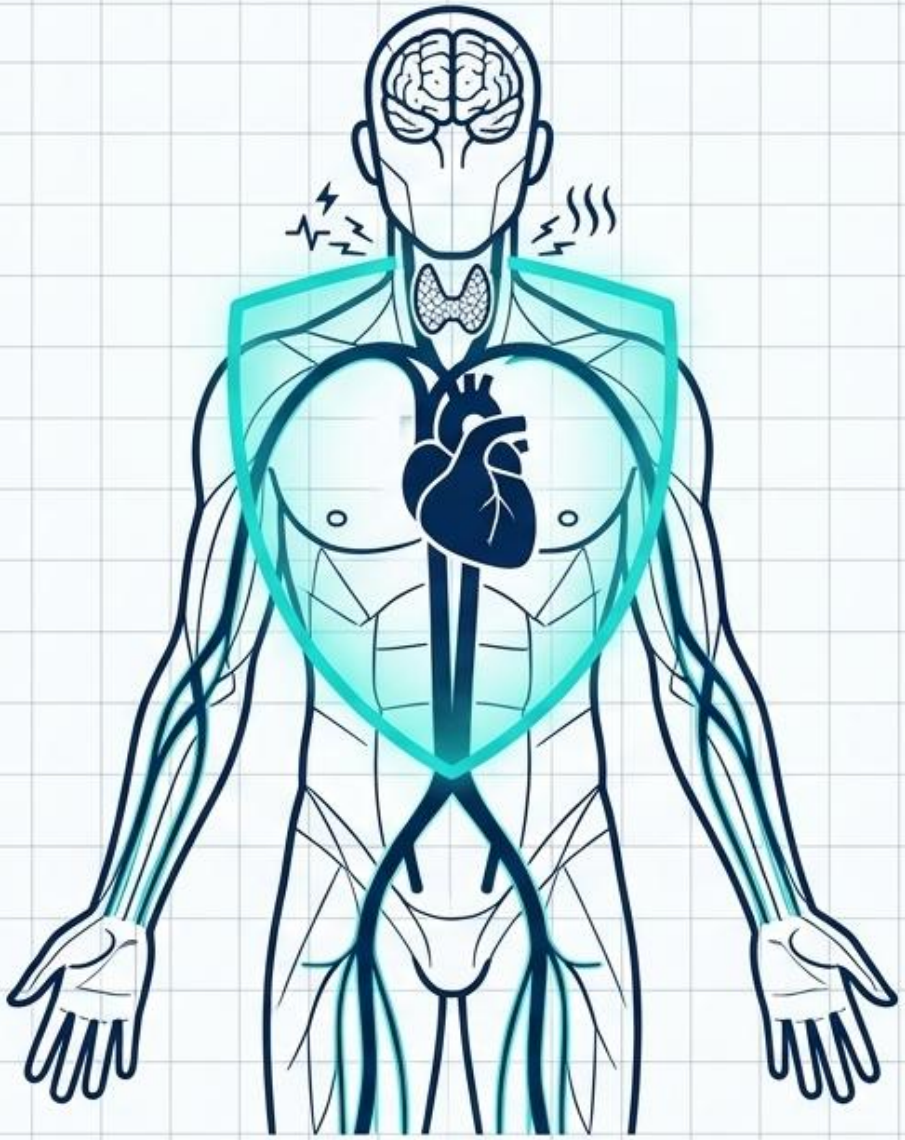
Pregnancy

Early pregnancy: PTU is preferred when antithyroid treatment is required.

After 16 weeks: individualize PTU versus MMI with specialist input. ATA 2026 specifies no preferred drug. Use the lowest effective dose.

Fever OR sore throat: stop drug and obtain urgent CBC.
Jaundice or dark urine: stop drug and seek urgent assessment.

ADJUVANTS SHIELD THE PERIPHERY FROM THYROTOXICOSIS



β-BLOCKERS (PROPRANOLOL)

Relieves tachycardia, tremor and anxiety.
High-dose propranolol also reduces T4-to-T3 conversion.
Caution: asthma, heart block and decompensated heart failure.

CALCIUM CHANNEL BLOCKERS (DILTIAZEM, VERAPAMIL)

Diltiazem can control heart rate when beta-blockers are unsuitable.
Avoid in reduced-ejection-fraction heart failure.

CORTICOSTEROIDS (DEXAMETHASONE, HYDROCORTISONE)

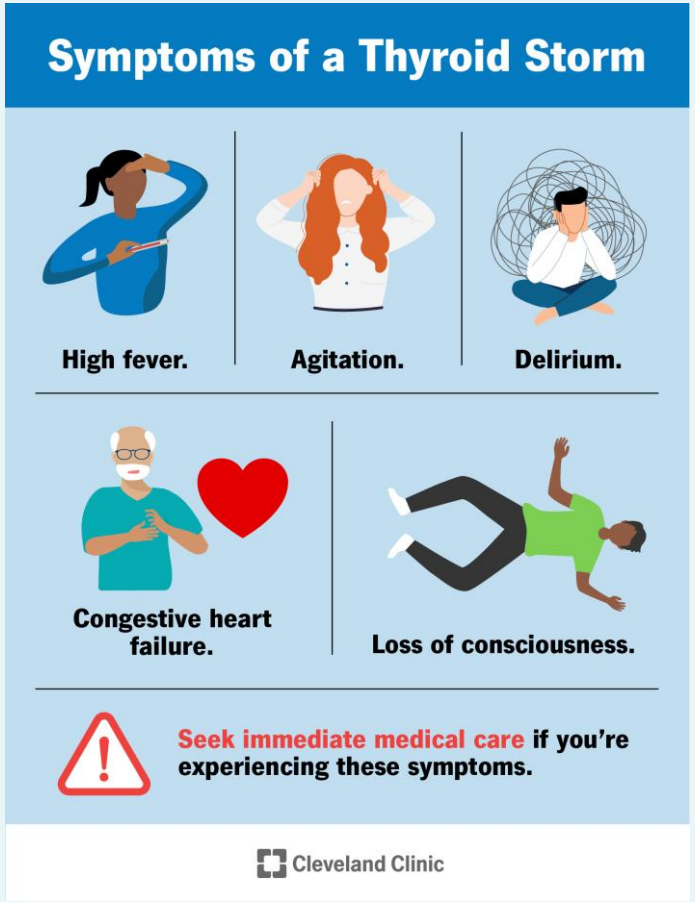
In thyroid storm: reduce T4-to-T3 conversion and provide adrenal support.
Specialist regimens also treat active Graves' orbitopathy.

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Thyroid storm: complementary drug actions

Fever, marked tachycardia and CNS or organ dysfunction require urgent hospital treatment.

Treatment	Purpose
Thioamide (PTU or MMI)	Reduce new hormone synthesis
Iodide after thioamide	Rapidly inhibit hormone release
Beta-blocker, if hemodynamically appropriate	Control adrenergic effects
Glucocorticoid	Reduce T4-to-T3 conversion and provide adrenal support
Supportive care and treatment of the trigger	Manage temperature, fluids and organ dysfunction



This is combined therapy in a monitored setting. Drug choice depends on liver and cardiac status.

Iodide: rapid, temporary inhibition

Examples

Potassium iodide (SSKI) and Lugol's solution provide nonradioactive iodide.

Main effects

High doses acutely inhibit thyroid hormone release and organification. They also reduce thyroid vascularity.

Clinical use

Short-term adjunct in thyroid storm and before thyroid surgery. Escape limits sustained efficacy.

Sequence and adverse effects

In storm, give after a thioamide, usually at least 1 hour later. Rash, GI upset and salivary swelling may occur.

Iodide can worsen hyperthyroidism in susceptible patients if used without appropriate preparation.

Radioactive iodine (I-131)

Mechanism

The thyroid concentrates iodine-131. Beta radiation damages follicular cells over weeks to months.

Clinical role

Definitive treatment for Graves’ disease or toxic nodules. It has no role in destructive thyroiditis.

Major limitations

Contraindicated during pregnancy and lactation. It can worsen Graves’ orbitopathy.

Follow-up

Hypothyroidism is common and may require lifelong T4. Follow thyroid tests and radiation-safety instructions.

Delay pregnancy for at least 6 months and until thyroid function is stable.